Grinding Mill



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Hammer Mill / Pulverizer; Stone Mill / Roller Mill

What it is used for

A grinding mill reduces suitable materials into smaller particles using a hammer, stone, roller, or other grinding mechanism. Select the correct mill setup for the material and follow the operating instructions to maintain product quality and safe operation.

Main parts

- Feed hopper and feed control
- Grinding chamber with hammers, stones, rollers, or grinding plates
- Drive motor and transmission
- Screens or particle-size adjustment mechanism
- Discharge outlet and collection container
- Protective guards and emergency stop, if fitted

Before you start

1. Identify the mill type and confirm that the material is suitable for processing.
2. Inspect the chamber, grinding elements, screens, belts, fasteners, and guards.
3. Check the motor, electrical connections, and emergency stop.
4. Ensure the mill and collection container are clean and dry as required.
5. Confirm the correct screen, roller gap, or grinding setting for the target particle size.
6. Remove metal, stones, and other foreign objects from the feed material.
7. Wear required PPE, including eye and hearing protection and suitable dust protection.

How to use

1. Keep the mill switched off and isolated while inspecting or changing components.
2. Fit the approved screen or set the grinding gap according to the product requirement.
3. Secure all covers, guards, and collection containers before starting.
4. Power on the mill according to the manufacturer's procedure.
5. Allow the machine to reach normal operating speed before feeding material, if specified.
6. Feed material gradually at the recommended rate without overloading the chamber.
7. Monitor motor sound, vibration, temperature, dust, and discharge flow.
8. Reduce or stop feeding if the mill becomes overloaded or output becomes irregular.
9. Collect the processed material in a suitable clean container.
10. Stop feeding and allow the remaining material to discharge according to the instructions.
11. Switch off and isolate the machine before opening covers or clearing a blockage.
12. Wait for all moving parts to stop completely before inspection or cleaning.

3. Remove remaining material and clean the machine using the approved method.
4. Record output, settings, and any abnormal noise, vibration, or faults.

⚠ Safety

- Only trained operators should operate the mill or change grinding components.
- Never insert hands or tools into the hopper or grinding chamber while the machine is connected to power.
- Keep all guards and covers fitted during operation.
- Isolate the power and verify complete stoppage before clearing blockages or maintenance.
- Control dust and use appropriate respiratory protection where the risk assessment requires it.
- Do not grind flammable, explosive, or hazardous materials unless the mill is specifically designed and approved for them.

Care & storage

- Clean the hopper, chamber, screens, and discharge path after use.
- Inspect hammers, stones, rollers, screens, bearings, and belts for wear.
- Lubricate bearings and service points according to the maintenance schedule.
- Check fasteners, guards, and electrical connections regularly.
- Keep the mill dry and store removable components safely.

Common mistakes to avoid

- Feeding material faster than the mill can process it.
- Using a damaged screen or incorrect grinding setting.
- Opening the chamber before the moving parts have stopped completely.
- Ignoring excessive dust, vibration, unusual noise, or overheating.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.